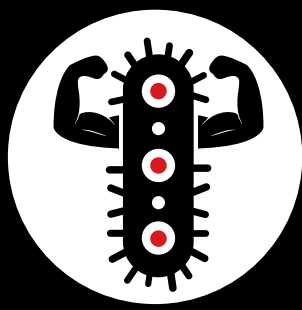
INDIA ON A SLIPPERY SLOPE



↘ **India is seeing a 5-10 per cent rise** in resistance to broad-spectrum antibiotics every year

↘ **Methicillin-resistant *Staphylococcus aureus*, which was virtually non-existent in India two decades ago, almost doubled from 28.4% to 42.6% between 2016 and 2021**

↘ **For diarrheagenic *E. coli* and *Shigella spp.*,** norfloxacin's efficacy has fallen drastically

↘ **The susceptibility of *E. Coli*,** which causes over 85 per cent of urinary tract infections (UTIs), to Imipenem dropped from 86 per cent in 2016 to 64 per cent in 2021

↘ For ***Acinetobacter baumannii*, resistance to** carbapenems stood at 87.5 per cent in 2021

(Source: ICMR; data collected from January to December 2021)

THE COST OF NEGLIGENCE

1.27
MILLION

The number of deaths caused directly by anti-microbial resistance in 2019 across the world, according to *The Lancet*

10
MILLION

The number of deaths per year that the WHO estimates AMR will cause by 2050

73.3
PER CENT

of blood infections and 53.1 per cent of UTIs were found to be antibiotic-resistant in a study conducted across 120 ICUs in the country by HAI-Surveillance India. Of these, 38.1 per cent of blood infection and 27.9 per cent of UTI cases had fatal outcomes within 14 days (HAIs, however, were a risk factor, not a direct cause)

surfaces of machines (if they are not sanitised well). But now, we have found a mutation in food-origin *Klebsiella* that allows it to convey the Colistin-resistant gene to *Klebsiella* inside humans," says Dr Ghafur, who led the study. Doctors say such indicators, while small in number at present, may mean that a large part of the population could soon become vulnerable. Drug-resistant bacteria can be present in your gut, spreading to others through contaminated surfaces. In the event of any hospitalisation or fall in immunity, these bugs then pose a huge risk to general health and recovery.

Separately, researchers have only recently started studying the impact of irresponsible antibiotic use in farming. A 2014 study conducted by the Centre for Science and Environment found that 40 per cent of the 70 chicken samples it analysed in Delhi tested positive for antibiotics. "The spread doesn't happen through eating cooked meat but by handling uncooked, contaminated meat. Resistant strains in animals can enter humans through direct contact with live or dead animals, exposure to contaminated surfaces, or through litter, which is often used as farming manure," says Dr Ghafur, adding that something as simple as touching uncooked chicken that has resistant bacteria and then touching our faces with contaminated hands can put us at risk.

Further, even though there has been a ban since 2019 on using Colistin for animals, doctors say other antibiotics continue to be used. The burgeoning fast food industry has also not made any public commitment to use antibiotic-free chicken in India, even as it has undertaken steps to cut down on the use of such meat in the West.

HOW DO BACTERIA DEVELOP RESISTANCE?

Recent studies have revealed fascinating insights into the mechanisms (see *How Bacteria Fight Back)* bacteria employ to overcome drugs. Once a mechanism works, the bacteria that have survived find ways to replicate and spread. For example, an alert issued in the UK in 2009 warned of an increasing number of carbapenem-resistant *Enterobacteriaceae* strains identified in hospital patients in the country. Many of them had been recently hospitalised in India and Pakistan. Later, an Indian study published in 2010 showed how the 'New Delhi Metallo-1', a resistance mechanism employed by *Enterobacteriaceae* for carbapenems, had spread to 22 people in just around three months at the Hinduja Hospital in Mumbai.

"DNA from a resistant germ can eventually alter the DNA of a non-resistant germ," says Dr Balaji. This can happen in various ways, and one among them is simply the power of survival. "Normally, our gut carries 10^{11} bacteria per gram of stool. But suppose you have a single resistant bacteria and you take the antibiotic it is resistant to, all the other bacteria will die, but the resistant ones will survive and grow to dominate your gut flora. They can also pass through your faeces and continue